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Unit-1 : MatricesSystem of Linear equation

An eqⁿ with variables $x_1, x_2, x_3, \dots, x_n$ is called a linear eqⁿ if the variables have a degree 1. The collection of all linear eqⁿs is called system of linear eqⁿ. The General form is:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

The Matrix form is

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$AX = B$$

where A is coefficient matrix, X is unknown matrix and B is constant matrix.

NOTE: ① If the system of eqⁿ is of the form $AX = 0$ (Matrix $B = 0$) is called Homogeneous SLE. Otherwise (Matrix $B \neq 0$) is called Non-HSLE.

② The 1st non-zero element of the row is called the leading entry or pivot. (LE).

③ The positions of pivot elements is called pivot position.

④ The columns in which pivot elements are present are pivot columns.

Elementary Transformation Matrix

1] ROW

- Interchanging 2 rows ($R_i \leftrightarrow R_j$)
- Multiplying all the entries in a row with non-zero constants ($R_j \rightarrow kR_j$)
- Replace 1 row by sum of itself and multiple of another row and a non-zero constant ($R_i \rightarrow R_i + kR_j$)

Echelon form of Matrix

We can say that given matrix is in echelon form if it satisfies following properties:

- The L.E of any row must be rightside of LE of previous row.
- The elements below the LE (in the column) must be 0.
- All the zero rows must be (the) at the bottom of the matrix

26] Rank of a Matrix

No. of Non zero rows in the echelon form of the Matrix denoted by $\rho(A)$.

Ex: $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 0 \end{bmatrix}$

, $\rho(A) = 2$

Summs

1. Find Rank.

$$a) A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} I. R_3 &\rightarrow R_3 + R_1 \\ R_2 &\rightarrow R_2 - 3R_1 \\ II. R_3 &\rightarrow R_3 + R_2 \end{aligned}$$

$$I. A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 3 & -10 \\ 0 & 2 & -3 & 10 \end{bmatrix}$$

$$II. A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 3 & -10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$b) A = \begin{bmatrix} 3 & -3 & 0 \\ 1 & 4 & 5 \\ 4 & 4 & 8 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} I. R_2 &\rightarrow 3R_2 - R_1 \\ R_3 &\rightarrow 3R_3 - 4R_1 \end{aligned}$$

$$A \sim \begin{bmatrix} 3 & -3 & 0 \\ 0 & 15 & 15 \\ 0 & 24 & 24 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 3 & -3 & 0 \\ 0 & 15 & 15 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \underline{\underline{P(A) = 2}}$$

$$c) A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

$$\begin{aligned} \text{I. } R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \\ R_4 &\rightarrow R_4 - 6R_1 \end{aligned}$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -8 & 3 \\ 0 & -4 & -11 & 5 \end{bmatrix}$$

$$\text{II. } R_2 \leftrightarrow R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -11 & 5 \end{bmatrix}$$

$$\text{III. } R_4 \rightarrow R_4 - R_2$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$$\text{IV. } R_4 \rightarrow R_4 - R_3$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = \underline{\underline{3}}$$

d) $A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 3 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$

I. $R_2 \rightarrow 2R_2 - R_1$
 $R_3 \rightarrow 2R_3 - R_1$

$A \sim \begin{bmatrix} 2 & -1 & -3 & -1 \\ 0 & 7 & 9 & -1 \\ 0 & 1 & 5 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix}$

II. $R_3 \rightarrow 7R_3 - R_2$
 $R_4 \rightarrow 7R_4 - R_2$

$A \sim \begin{bmatrix} 2 & -1 & -3 & -1 \\ 0 & 7 & 9 & -1 \\ 0 & 0 & 26 & 22 \\ 0 & 0 & -2 & -6 \end{bmatrix}$

III. $R_4 \rightarrow 13R_4 + R_3$

$A \sim \begin{bmatrix} 2 & -1 & -3 & -1 \\ 0 & 7 & 9 & -1 \\ 0 & 0 & 26 & 22 \\ 0 & 0 & 0 & -58 \end{bmatrix}$

$\therefore \rho(A) = 4$

$A = (A|I)$

c) $A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \end{bmatrix}$

I. $R_2 \rightarrow 2R_2 - R_1$
 $R_4 \rightarrow 2R_4 - R_1$

$A \sim \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & -1 & 2 & 1 \end{bmatrix}$

II. $R_3 \rightarrow 3R_3 - R_2$
 $R_4 \rightarrow 3R_4 + R_2$

$A \sim \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & 12 & 12 \end{bmatrix}$

III. $R_4 \rightarrow R_4 - 3R_3$

$A \sim \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix}$

$\therefore \underline{\underline{p(A) = 4}}$

f) A

f) $A = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 3 & 1 \\ 1 & 0 & 1 & 3 \end{bmatrix}$

$R_2 \rightarrow 3R_2 - R_1$

$R_4 \rightarrow 3R_4 - R_1$

$\sim \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 8 & 3 & 0 \\ 0 & 1 & 3 & 1 \\ 0 & -1 & 3 & 9 \end{bmatrix}$

$R_3 \rightarrow 8R_3 - R_2$; $R_4 \rightarrow 8R_4 + R_2$

$\sim \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 8 & 3 & 0 \\ 0 & 0 & 21 & 8 \\ 0 & 0 & 27 & 92 \end{bmatrix}$

$R_4 \rightarrow 27R_4 - 21R_3$

$\sim \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 8 & 3 & 0 \\ 0 & 0 & 21 & 8 \\ 0 & 0 & 0 & 1296 \end{bmatrix}$

$\rho(A) = 4$

8/1 Consistent and Inconsistent system of LE

Consider a sys. of eqⁿ (or) $AX=B$, it is consistent if it has a solⁿ otherwise it is inconsistent.

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$$\text{let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

A matrix is formed by appending to the matrix A by an extra column consisting of the elements of B is called augment matrix. It is denoted by $[A : B]$ and given by

$$[A : B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & : & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & : & b_2 \\ \vdots & \vdots & \ddots & \vdots & : & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & : & b_m \end{bmatrix}$$

If the sys. of eqⁿ are consistent, then there are soln.

(i) Unique solⁿ $\rightarrow P(A) = P(A : B) = n = n$
 (ii) Infinite solⁿ $\rightarrow P(A) = P(A : B) = r < n$

where n is no. of unknown variables in the given sys. of eqⁿs.

And the sys. is inconsistent, if there is no solⁿ i.e.

$$P(A) \neq P(A : B)$$

NOTE: The variables corresponding to the pivot column are called basic variables and others are called free "

Q. Sum

Test for consistency & solve.

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$$1: 2x_1 - x_2 + 3x_3 = 1 ; -3x_1 + 4x_2 - 5x_3 = 0 ; x_1 + 3x_2 - 6x_3 = 0$$

$$\Rightarrow [A:B] = \begin{bmatrix} 2 & -1 & 3 & : & 1 \\ -3 & 4 & -5 & : & 0 \\ 1 & 3 & -6 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow 2R_2 + 3R_1 ; R_3 \rightarrow 2R_3 - R_1$$

$$\sim \begin{bmatrix} 2 & -1 & 3 & : & 1 \\ 0 & 5 & -1 & : & 3 \\ 0 & 7 & -15 & : & -1 \end{bmatrix}$$

$$R_3 \rightarrow 5R_3 - 7R_2$$

$$\sim \begin{bmatrix} 2 & -1 & 3 & : & 1 \\ 0 & 5 & -1 & : & 3 \\ 0 & 0 & -68 & : & -26 \end{bmatrix}$$

$$P(A) = 3 ; P(A:B) = 3 ; n = 3$$

$$\therefore P(A) = P(\underline{A:B}) = n = 3$$

This system is consistent and has unique solⁿ.

$$AX = B$$

$$X = (A^{-1}B) ; C = (B:A) ; S = (A) ; T = (A)^{-1}$$

$$2x_1 - x_2 + 3x_3 = 1 \quad \text{--- ①}$$

$$0 + 5x_2 - x_3 = 3 \quad \text{--- ②}$$

$$0 + 0 - 68x_3 = -26 \quad \text{--- ③}$$

$$\text{From ③, } x_3 = \frac{13}{34}, \text{ From ②, } x_2 = \frac{115}{170} = \frac{23}{34}$$

$$\text{From ①, } x_1 = \frac{9}{34}$$

2. $x + 2y + 2z = 1$
 $2x + y + z = 2$
 $3x + 2y + 2z = 3$
 $y + z = 0$

$$\Rightarrow [A:B] = \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ 3 & 2 & 2 & 3 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1 ; R_3 \rightarrow R_3 - 3R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -3 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

$$R_3 \rightarrow 3R_3 - 4R_2 ; R_4 \rightarrow 3R_4 + R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 2 & 1 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$P(A) = 2 ; P(A:B) = 2 ; n = 3$$

$$P(A) = \underline{P(A:B)} \neq n$$

It is consistent and has ∞ solⁿ.

* The pivot elements present in 1st & 2nd column.
 x & y are basic and z is free variable

$$x + 2y + 2z = 1 ; -3y - 3z = 0 \Rightarrow y + z = 0$$

—①

Since z is a free variable, it has ∞ values, the values of x & y depend on z .

From ②, $-z = +y$

From ①, $x = 1$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -z \\ z \end{bmatrix}$$

9/1 Solution using Gauss elimination method

Procedure ① ^{write} The Augmented matrix of given sys.

② Transform the Augmented matrix to regular form (The echelon form matrix will be a Upper Δ matrix)

③ Write the matrix into linear eq's

④ By back substitution, obtain the unknowns.

SUMS (using Gauss elimination)

1. $x + y + z = 9$; $2x - 3y + 4z = 13$; $3x + 4y + 5z = 40$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \\ 3 & 4 & 5 \end{bmatrix} = A \quad \begin{bmatrix} 9 \\ 13 \\ 40 \end{bmatrix} = B$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & -3 & 4 & 13 \\ 3 & 4 & 5 & 40 \end{array} \right]$$

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$$R_2 \rightarrow R_2 - 2R_1 ; R_3 \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -5 & 2 & : & -5 \\ 0 & 1 & 2 & : & 13 \end{bmatrix}$$

$$R_3 \rightarrow 5R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 9 \\ 0 & -5 & 2 & : & -5 \\ 0 & 0 & 12 & : & 60 \end{bmatrix}$$

$$P(A) = 3 ; P(A:B) = 3$$

$$n = 3$$

$$P(A) = P(A:B) = n = 3$$

Unique solⁿ:

$$x + y + z = 9 \quad \text{--- ①}$$

$$-5y + 2z = -5 \quad \text{--- ②}$$

$$12z = 60 \quad \text{--- ③}$$

From ③, $z = 5$

From ②, $y = 3$

From ①, $x = 1$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

$$2. 5x + 3y + 7z = 4 ; 3x + 26y + 2z = 9 ; 7x + 2y + 10z = 5$$

$$\Rightarrow [A:B] = \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 3 & 26 & 2 & : & 9 \\ 7 & 2 & 10 & : & 5 \end{bmatrix}$$

$$R_2 \rightarrow 5R_2 - 3R_1 ; R_3 \rightarrow 5R_3 - 7R_1$$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 12 & -11 & : & 33 \\ 0 & -11 & 1 & : & -3 \end{bmatrix}$$

$$R_3 \rightarrow 11R_3 + R_2$$

$$\sim \begin{bmatrix} 5 & 3 & 7 & : & 4 \\ 0 & 12 & -11 & : & 33 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$P(A) = P(A|B) = 2$$

$$n = 3$$

Infinite solⁿ

$$5x + 3y + 7z = 4 \quad \text{--- ①}$$

$$12y - 11z = 33 \quad \text{--- ②}$$

From ②

$$11y - z = 3$$

$$y = \frac{3+z}{11}$$

From ①

$$5x + 3\left(\frac{3+z}{11}\right) + 7z = 4$$

$$5x = -\frac{9+3z}{11} - 7z + 4$$

$$5x = -\frac{9+3z}{11} - 7z + 4$$

$$x = -\frac{9+3z}{55} - \frac{7z}{5} + \frac{4}{5}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (-1-16z)/11 \\ (3+z)/11 \\ z \end{bmatrix}$$

3/5

3. $x+4y-z=5$; $x+y-6z=-12$; $3x-y-z=4$

$$\Rightarrow [A:B] = \begin{bmatrix} 1 & 4 & -1 & : & -5 \\ 1 & 1 & -6 & : & -12 \\ 3 & -1 & -1 & : & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 ; R_3 \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 4 & -1 & : & -5 \\ 0 & -3 & -5 & : & -7 \\ 0 & -13 & 2 & : & 19 \end{bmatrix}$$

$$R_3 \rightarrow 3R_3 - 13R_2$$

$$\sim \begin{bmatrix} 1 & 4 & -1 & : & -5 \\ 0 & -3 & -5 & : & -7 \\ 0 & 0 & 71 & : & 148 \end{bmatrix}$$

$$P(A) = P(A:B) = n = 3$$

Unique solⁿ

$$x+4y-z = -5 \quad \text{--- ①}$$

$$-3y-5z = -7 \quad \text{--- ②}$$

$$71z = 148 \quad \text{--- ③}$$

From ③, $z = 148/71$

From ②, $y = -81/71$

From ①, $x = 117/71$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 117/71 \\ -81/71 \\ 148/71 \end{bmatrix}$$

$$y = \frac{-7+5z}{-3}$$

$$y = \frac{-7+5(148)}{71}$$

$$-3$$

$$= -497 + 740$$

$$= 243$$

$$= -243$$

$$243$$

Q Determine the value of a & b , so the system of eqⁿs

$$x + 2y + 3z = 6 ; x + 3y + 5z = 9 ; 2x + 5y + az = b$$

has (i) No (ii) Unique (iii) ∞

$$\Rightarrow [A:B] = \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 1 & 3 & 5 & : & 9 \\ 2 & 5 & a & : & b \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 ; R_3 \rightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 0 & 1 & 2 & : & 3 \\ 0 & 1 & a-6 & : & b-12 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & : & 6 \\ 0 & 1 & 2 & : & 3 \\ 0 & 0 & a-8 & : & b-15 \end{bmatrix}$$

- ① $a = 8 ; b \neq 15$
- ② $a \neq 8 ; b$ any value
- ③ $a = 8 ; b = 15$

5. $2x + y + z = 10 ; 3x + 2y + 3z = 18 ; x + 4y + 9z = 16$

$$\Rightarrow [A:B] = \begin{bmatrix} 2 & 1 & 1 & : & 10 \\ 3 & 2 & 3 & : & 18 \\ 1 & 4 & 9 & : & 16 \end{bmatrix}$$

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$$[A:B] = \begin{bmatrix} 2 & 1 & 1 & : & 10 \\ 3 & 2 & 3 & : & 18 \\ 1 & 4 & 9 & : & 16 \end{bmatrix}$$

$$R_2 \rightarrow 2R_2 - 3R_1 ; R_3 \rightarrow 2R_3 - R_1$$

$$\sim \begin{bmatrix} 2 & 1 & 1 & : & 10 \\ 0 & 1 & 3 & : & 6 \\ 0 & 7 & 17 & : & 22 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 7R_2$$

$$\sim \begin{bmatrix} 2 & 1 & 1 & : & 10 \\ 0 & 1 & 3 & : & 6 \\ 0 & 0 & -4 & : & -20 \end{bmatrix}$$

$$P(A) = P(A:B) = n = 3$$

Unique Solⁿ

$$\begin{aligned} 2x + y + z &= 10 \quad \text{--- ①} \\ y + 3z &= 6 \quad \text{--- ②} \\ y - 4z &= -20 \quad \text{--- ③} \end{aligned}$$

From ③

$$\underline{z = 5}$$

From ②

$$\underline{y = -9}$$

From ①

$$\underline{x = 7}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ -9 \\ 5 \end{bmatrix}$$

6- Determine a ~~value~~ such that $x+y-z=1$; $2x+3y+az=3$; $x+ay+3z=2$ ① No ② ∞ ③ Unique

$$\Rightarrow [A:B] = \begin{bmatrix} 1 & 1 & -1 & : & 1 \\ 2 & 3 & a & : & 3 \\ 1 & a & 3 & : & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad ; \quad R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & -1 & : & 1 \\ 0 & 1 & a+2 & : & 1 \\ 0 & a-1 & 4 & : & 1 \end{bmatrix}$$

$$R_3 \leftrightarrow R_2 \quad ; \quad R_3 \rightarrow R_3 - (a-1)R_2$$

$$\sim \begin{bmatrix} 1 & 1 & -1 & : & 1 \\ 0 & a-1 & 4 & : & 1 \\ 0 & 1 & a+2 & : & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & -1 & : & 1 \\ 0 & a-1 & a+2 & : & 1 \\ 0 & 0 & 6-a^2-a & : & 2-a \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & -1 & : & 1 \\ 0 & 1 & a+2 & : & 1 \\ 0 & 0 & (a+3)(a-2) & : & 2-a \end{bmatrix}$$

① No : $a = -3$; ② ∞ : $a = 2$; ③ Unique : $a \neq -3, 2$

L U Decomposition

is an eqⁿ that express the Matrix A as product of 2 matrices.

Let A be a $m \times n$ matrix that can be reduced to echelon form without row interchange.

The matrix A can be written as $A = LU$ where L is lower Δ matrix with diagonal as 1 & U is upper Δ matrix.

i.e. $A = LU$

$$= \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Find L & U for A.

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + R_1; R_3 \rightarrow R_3 - 2R_1; R_4 \rightarrow R_4 + 3R_1$$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 10 & 4 & -9 \\ 0 & -16 & -11 & 18 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 5R_2; R_4 \rightarrow R_4 - 8R_2$$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 3R_3$$

$$\sim \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} = U$$

I. Consider a sq. matrix A transforming to echelon form, the obtained echelon matrix is U .

II. In each step, before making the elements which are below the leading entry zero, divide the leading entry replace the same position in L by resultant elements.

Example: Find L & U .

$$1. A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & 2 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A = LU \\ = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -5 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$2. A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & -1 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2 \quad \sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{bmatrix} = U$$

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$$A = LU$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix}$$

$$3. A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 + 3R_1$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 3R_2$$

$$R_4 \rightarrow R_4 - 4R_2$$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 2R_3$$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = U$$

$$A = LU$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

$$4. A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 6 & 5 & 2 \\ 8 & 7 & 3 \\ 10 & 9 & 4 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 4 & 3 & 0 & 1 & 0 \\ 5 & 4 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 ; R_4 \rightarrow R_4 - 4R_1 \\ R_3 \rightarrow R_3 - 3R_1 ; R_5 \rightarrow R_5 - 5R_1$$

$$\sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 3 & 3 \\ 0 & 4 & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2 ; R_4 \rightarrow R_4 - 3R_2 \\ R_5 \rightarrow R_5 - 4R_2$$

$$\sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U$$

$$A = LU$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 \\ 4 & 3 & 0 & 1 & 0 \\ 5 & 4 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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Eigen values & vectors

Let A be a sq. matrix of order n & eigen vector of A is a non-0 vector such that $AX = \lambda X$ for some scalar λ where λ is called eigen value of matrix A . If there is a non-trivial solⁿ $AX = \lambda X$, the vector X is called eigen vector corresponding to eigen value λ .

Consider a unit sq. matrix of order n , then $|A - \lambda I| = 0$ gives the characteristic eqⁿ, the roots of this eqⁿ are eigen values. We can find eigen vector corresponding eigen values using $(A - \lambda I)X = 0$.

NOTE: ① The sum of eigen values = Sum of the elements of main diagonal.

② $\text{Tr}(\text{sq. matrix})$ is sum of its main diagonal elements.
i.e. $\text{Tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$.

③ The characteristic eqⁿ is a) For 2×2 : $\lambda^2 - \text{Tr}(A)\lambda + |A| = 0$

b) For 3×3 : $\lambda^3 - \text{Tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$
where M_{ii} is the principal minor

Ex: Find eigen value & vector of $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Step 1: $A - \lambda I$

$$\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \Rightarrow \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix}$$

Step 2: $|A - \lambda I| = 0$

$$\begin{aligned}(5-\lambda)(2-\lambda) - 4 &= 0 \\ 10 - 2\lambda - 5\lambda + \lambda^2 - 4 &= 0 \\ -7\lambda + \lambda^2 + 6 &= 0 \\ \lambda^2 - 7\lambda + 6 &= 0 \\ (\lambda - 6)(\lambda - 1) &= 0 \\ \lambda &= \underline{\underline{6, 1}}\end{aligned}$$

Step 3: Characteristic eqⁿ

$$\begin{aligned}\lambda^2 - \text{Tr}(A)\lambda + |A| &= 0 \\ \lambda^2 - 7\lambda + 6 &= 0\end{aligned}$$

Step 4: Find $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ when $\lambda = 6$ using $(A - \lambda I)x = 0$

$$A - \lambda I = \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -x_1 + 4x_2 \\ x_1 - 4x_2 \end{bmatrix} = 0$$

$$-x_1 + 4x_2 = 0 \Rightarrow x_1 = 4x_2$$

$$x_1 - 4x_2 = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\lambda = \underline{\underline{1}}$$

$$A - \lambda I = \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 4x_1 + 4x_2 \\ x_1 + x_2 \end{bmatrix} = 0$$

$$\Rightarrow x_1 = -x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Sums

$$\textcircled{1} \begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix}$$

$$|A - \lambda I| \Rightarrow \begin{bmatrix} 4 & -5 \\ 1 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 4-\lambda & -5 \\ 1 & -2-\lambda \end{bmatrix}$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0$$

$$(\lambda - 3)(\lambda + 1) = 0$$

$$\lambda = 3, -1$$

$$(A - \lambda I)x = 0 \text{ when } \lambda = 3$$

$$\begin{bmatrix} 4-3 & -5 \\ 1 & -2-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_1 - 5x_2 \\ x_1 - 5x_2 \end{bmatrix} = 0 \Rightarrow x_1 = 5x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

when $\lambda = 1$

$$\begin{bmatrix} 4+1 & -5 \\ 1 & -2+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 5x_1 - 5x_2 \\ x_1 - x_2 \end{bmatrix} = 0$$

$$\begin{aligned} 3x_1 - 5x_2 &= 0 \\ 3x_1 - 9x_2 &= 0 \end{aligned} \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\textcircled{2} \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

$$\begin{aligned} \lambda^2 + 7\lambda + 6 &= 0 \\ (\lambda + 6)(\lambda + 1) &= 0 \\ \lambda &= -6, -1 \end{aligned}$$

$$\lambda = -6$$

$$\lambda = -1$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + 2x_2 = 0$$

$$x_1 = -2x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-4x_1 + 2x_2 = 0$$

$$2x_1 = x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$(3) \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

$$A - \lambda I \Rightarrow \begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix}$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$= (\lambda - 3)(\lambda - 2)(\lambda - 1) = 0$$

$$\lambda = \underline{1, 2, 3}$$

$$\lambda = 1$$

$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$+x_3 = 0 \Rightarrow x_3 = 0$$

$$\begin{bmatrix} -x_3 \\ x_1 + x_2 + x_3 \\ 2x_1 + 2x_2 + 2x_3 \end{bmatrix} = 0$$

$$\Rightarrow x_3 = 0$$

$$x_1 = -x_2$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda = 2$$

$$+ y \quad z$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -x_1 - x_3 \\ x_1 + x_3 \\ 2x_1 + 2x_2 + x_3 \end{bmatrix} = 0$$

$$x_1 = -x_3$$

$$3x_1 + 2x_2 = 0$$

$$x_1 = -\frac{2x_2}{3}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ +3 \\ +1 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -2x_1 - x_3 \\ x_1 - x_2 + x_3 \\ 2x_1 + 2x_2 \end{bmatrix} = 0$$

$$x_1 = -x_2$$

$$x_1 = \frac{-x_3}{2}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

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Cramer's Rule

Consider any 2 eqns

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

$$\Rightarrow x = \frac{-(b_1c_2 - b_2c_1)}{a_1c_2 - a_2c_1} = k$$

$$\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \quad \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} \quad \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

④ $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 2 & 2 & 3 \end{bmatrix}$

$$A - \lambda I \Rightarrow \begin{bmatrix} 1-\lambda & 0 & 0 \\ 1 & 2-\lambda & 0 \\ 2 & 2 & 3-\lambda \end{bmatrix}$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\lambda = 1, 3, 2$$

$$\lambda = 1$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 2 \end{bmatrix} \Rightarrow \begin{aligned} x + y + z &= 0 \\ 2x + 2y + 2z &= 0 \end{aligned}$$

$$\frac{x}{b_1 c_1} = \frac{-y}{a_1 c_1} = \frac{z}{a_2 b_2} = k$$

$$\frac{x}{2} = \frac{-y}{0} = \frac{z}{0} = k$$

$$(x, y, z) = (2k, -0k, 0k) \Rightarrow x=2, y=0, z=0$$

$$\lambda = 2$$

$$\begin{bmatrix} -1 & 0 & 0 \\ +1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{aligned} -x + y + z &= 0 \\ +x + y + z &= 0 \\ 2x + 2y + z &= 0 \end{aligned}$$

$$\frac{x}{\begin{vmatrix} 0 & 0 \\ 2 & 1 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} +1 & 0 \\ 2 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} +1 & 0 \\ 2 & 2 \end{vmatrix}} = k$$

$$(x, y, z) = (0k, -1k, +2k) \Rightarrow x=0, \underline{y=-1}, \underline{z=2}$$

$$\lambda = 3$$

$$\begin{bmatrix} -2 & 0 & 0 \\ 1 & -1 & 0 \\ 2 & 2 & 0 \end{bmatrix} \Rightarrow \begin{aligned} -2x &= 0 \\ x - y &= 0 \\ 2x + 2y &= 0 \end{aligned}$$

$$\frac{x}{\begin{vmatrix} -1 & 0 \\ 2 & 0 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 1 & 0 \\ 2 & 0 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}} = k$$

$$(x, y, z) = (0k, 0k, 4k) \Rightarrow x=0, \underline{y=0}, \underline{z=4}$$

$$⑤ \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{bmatrix}$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$$

$$\lambda^3 - 6\lambda^2 + \underset{9}{\cancel{9}}\lambda - 4 = 0$$

$$\underline{\underline{\lambda = 4, 1, 1}}$$

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$$\lambda = 1$$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x - y + z = 0 \\ -x + y - z = 0 \\ x - y + z = 0 \end{cases}$$

Since there are 2 D.F., there will be 2 free variables, so cannot apply Cramer's rule

Let x, y be the free variables, $z = y - x$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ y - x \end{bmatrix} \Rightarrow x \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore \text{Eigen values} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda = 4$$

$$\begin{bmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -2x - y + z = 0 \\ -x - 2y - z = 0 \\ x - y - 2z = 0 \end{cases}$$

$$x = -y = z = k$$

$$\begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix} \quad \begin{bmatrix} -1 & -1 \\ 1 & -2 \end{bmatrix} \quad \begin{bmatrix} -1 & -2 \\ 1 & -1 \end{bmatrix}$$

$$\frac{x}{3} = \frac{-y}{3} = \frac{z}{3} = k \Rightarrow (x, y, z) = (3k, -3k, 3k)$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$A = \lambda I$$

$$⑥ \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} 3-\lambda & -2 & 4 \\ -2 & 6-\lambda & 2 \\ 4 & 2 & 3-\lambda \end{bmatrix}$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$$

$$\lambda^3 - 12\lambda^2 + 21\lambda + \frac{102}{98} = 0$$

$$\lambda = \underline{\underline{-2}}, 7, 7$$

$$\lambda = -2$$

$$\begin{bmatrix} 5 & -2 & 4 \\ -2 & 8 & 2 \\ 4 & 2 & 5 \end{bmatrix}$$

$$\frac{x}{36} = \frac{-y}{-18} = \frac{z}{-36} = k$$

$$x, y, z = k(2, 1, -2)$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$\lambda = 7$$

$$\begin{bmatrix} -4 & -2 & 4 \\ -2 & -1 & 2 \\ 4 & 2 & -4 \end{bmatrix}$$

$$\frac{x}{0} = \frac{-y}{0} = \frac{z}{0} = k$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ (2x+y)/2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1/2 \end{bmatrix}$$

$$\therefore \text{Eigen values} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1/2 \end{bmatrix}$$

$$\textcircled{7} \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} = A - \lambda I$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + (M_{11} + M_{22} + M_{33})\lambda - |A| = 0$$

$$\lambda^3 - 10\lambda^2 + (31)\lambda - 30 = 0$$

$$\lambda = 5, 3, 2$$

$$\lambda = 5$$

$$\begin{bmatrix} -2 & 1 & 4 \\ 0 & -3 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{x}{18} = \frac{-y}{-12} = \frac{z}{6} = k$$

$$(x, y, z) = 6k(3, 2, 1)$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} 0 & 1 & 4 \\ 0 & -1 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\frac{x}{10} = \frac{-y}{0} = \frac{z}{0} = k$$

$$(x, y, z) = 10k(-1, 0, 0)$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda = 2$$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow \frac{x}{6} = \frac{-y}{6} = \frac{z}{0} = k$$

$$(x, y, z) = 6k(1, -1, 0)$$

$$\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

22/1/26 Characteristic polynomial and minimal polynomial

A characteristic polynomial^(CP) of a sq. matrix A is defined as

$$X(\lambda) = |A - \lambda I| \text{, where } I \text{ is identity matrix of order same as } A.$$

A minimal polynomial^(MP) is a monic polynomial of lowest degree such that $m(A) = 0$ and it divides the characteristic polynomial.

Properties of M.P.

- Every sq. matrix has unique MP.
- M.P. of A divides characteristic polynomial.
- The pairs of factors $m(\lambda)$ are $\leq$ to those in CP.
- If the eigen values are distinct then $m(\lambda) = \text{CP}$.

Procedure

- Find the characteristic polynomial of a given matrix.
- List out all the possible MP.
- Substitute A in each candidate polynomial starting from lowest degree.
- The 1st MP for which $m(A) = 0$ is the MP for the given matrix.

Exs

Find CP & MP

$$1. \begin{bmatrix} 5 & 1 \\ 3 & 7 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} 5-\lambda & 1 \\ 3 & 7-\lambda \end{bmatrix}$$

$$\lambda^2 - \text{tr}(A)\lambda + |A| = 0$$

$$\lambda^2 - 12\lambda + 32 = 0$$

$$(\lambda - 8)(\lambda - 4) = 0$$

$$m(\lambda) = (\lambda - 8)(\lambda - 4)$$

$$m(A) = 0$$

$$m(A) = (A - \lambda_1 I)(A - \lambda_2 I)$$

$$(A - 8I)(A - 4I) = 0$$

$$\begin{bmatrix} -3 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

$$\therefore M.P. = (\lambda - 8)(\lambda - 4)$$

$$2. \begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} 4-\lambda & -1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$\lambda^2 - 6\lambda + 9 = 0$$

$$(\lambda - 3)(\lambda - 3)$$

$$m(A) = 0$$

$$(A - 3I) \Rightarrow \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \neq 0$$

$$(A - 3I)^2 \Rightarrow \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \underline{\underline{0}}$$

$$\therefore \text{M.P.} = (\lambda - 3)^2$$

$$3. \begin{bmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} 2-\lambda & 2 & -5 \\ 3 & 7-\lambda & -15 \\ 1 & 2 & -4-\lambda \end{bmatrix} \therefore$$

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

$$(\lambda - 3)(\lambda - 1)^2$$

$$m(\lambda) = (\lambda - 3)(\lambda - 1)$$

$$m(\lambda) = (\lambda - 3)(\lambda - 1)^2$$

$$m(A) = (A - 3I)(A - I)$$

$$m(A) = (A - 3I)(A - I)(A - I)$$

$$\begin{bmatrix} -1 & 2 & -5 \\ 3 & 4 & -15 \\ 1 & 2 & -7 \end{bmatrix} \begin{bmatrix} 1 & 2 & -5 \\ 3 & 6 & -15 \\ 1 & 2 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & -5 \\ 3 & 6 & -15 \\ 1 & 2 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \underline{\underline{0}}$$

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$$\therefore \text{M.P.} = (\lambda-3)(\lambda-1)$$

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 3 & 8 & -3 \\ 3 & 6 & -1 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} 3-\lambda & 2 & -1 \\ 3 & 8-\lambda & -3 \\ 3 & 6 & -1-\lambda \end{bmatrix}$$

$$\lambda^3 - 10\lambda^2 + 28\lambda - 24 = 0$$

$$(\lambda-2)^2(\lambda-6)$$

$$m(\lambda) = (\lambda-2)(\lambda-6)$$

$$m(\lambda) = (\lambda-2)^2(\lambda-6)$$

$$m(A) = (A-2I)(A-6I)$$

$$m(A) = (A-2I)(A-2I)(A-6I)$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 3 & 6 & -3 \end{bmatrix} \begin{bmatrix} -3 & 2 & -1 \\ 3 & 2 & -3 \\ 3 & 6 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 3 & 6 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \underline{\underline{0}}$$

$$\text{M.P.} = (\lambda-2)(\lambda-6)$$

26 Rayleigh's power method

It is an iterative method to compute approx. value of the dominant eigenvalue (with largest magnitude) & its corresponding eigen

vector for a given matrix.

Procedure: ① Consider matrix A and choose 1 initial non-zero vector say $X_0 = [x_1 \ x_2 \ x_3]^T$

② Compute AX_0 . Equate $AX_0 = \lambda_1 X_1$ where λ_1 is the largest abs value in AX_0 .

③ Compute $AX_1 = \lambda_2 X_2$. Repeat it till we get desired accurate λ value.

Sums

Q. Find dominant eigen value & vector. Carry out 5 iterations.

$$1. \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} = A ; X_0 = [1 \ 0 \ 0]^T$$

$$AX_0 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \lambda_1 X_1$$

$$AX_1 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 2 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \lambda_2 X_2$$

$$AX_2 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 0 \\ 2.6 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ 0 \\ 0.9285 \end{bmatrix} = \lambda_3 X_3$$

$$AX_3 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.9285 \end{bmatrix} = \begin{bmatrix} 2.9285 \\ 0 \\ 2.857 \end{bmatrix} = 2.9285 \begin{bmatrix} 1 \\ 0 \\ 0.9285 \end{bmatrix} = \lambda_4 X_4$$

$$AX_4 = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.9755 \end{bmatrix} = \begin{bmatrix} 2.9755 \\ 0 \\ 2.951 \end{bmatrix} = 2.9755 \begin{bmatrix} 1 \\ 0 \\ 0.9918 \end{bmatrix}$$

$$2. \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} = A; \quad X_5 = \begin{bmatrix} 1 \\ 0.8 \\ 0.8 \end{bmatrix}$$

$$AX = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0.8 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 4 \\ 3.6 \\ 2.8 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0.5294 \\ 0.7 \end{bmatrix}$$

$$AX_1 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1.8823 \\ 1.7647 \\ 6.7058 \end{bmatrix} = 6.7058 \begin{bmatrix} 0.2807 \\ 0.2631 \\ 1 \end{bmatrix}$$

$$AX_2 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} 0.2807 \\ 0.2631 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.3859 \\ 0.3508 \\ 5.8246 \end{bmatrix} = 5.8246 \begin{bmatrix} 0.0662 \\ 0.0602 \\ 1 \end{bmatrix}$$

$$AX_3 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} -0.1299 \\ -0.1322 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.6747 \\ -0.6867 \\ 5.1928 \end{bmatrix} = 5.1928 \begin{bmatrix} -0.1299 \\ -0.1322 \\ 1 \end{bmatrix}$$

$$AX_4 = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} -0.1299 \\ -0.1322 \\ 1 \end{bmatrix} = \begin{bmatrix} -1.6519 \\ -1.6566 \\ 4.6078 \end{bmatrix} = 4.6078 \begin{bmatrix} -0.3585 \\ -0.3599 \\ 1 \end{bmatrix}$$

$$AX_5 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$3. \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$4. \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$3 \Rightarrow \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1/3 \\ 1 \\ -1/3 \end{bmatrix} = 3 \begin{bmatrix} 0.3333 \\ 1 \\ -0.3333 \end{bmatrix}$$

$$AX_1 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.3333 \\ 1 \\ -0.3333 \end{bmatrix} = \begin{bmatrix} 3.6666 \\ 1.6667 \\ 0.3333 \end{bmatrix} - 3.6666 \begin{bmatrix} 0.3333 \\ 1 \\ -0.3333 \end{bmatrix}$$

$$AX_2 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0.7575 \\ -0.091 \end{bmatrix} = \begin{bmatrix} 2.2726 \\ 4.2731 \\ 1.7283 \end{bmatrix} = 4.2731 \begin{bmatrix} 0.5318 \\ 1 \\ 0.4044 \end{bmatrix}$$

$$AX_3 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.5318 \\ 1 \\ 0.4044 \end{bmatrix} = \begin{bmatrix} 3.1274 \\ 5.2134 \\ 7.5129 \end{bmatrix} = 7.5129 \begin{bmatrix} 0.4162 \\ 0.6939 \\ 1 \end{bmatrix}$$

$$AX_4 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.4162 \\ 0.6939 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.498 \\ 6.6367 \\ 12.359 \end{bmatrix} = 12.359 \begin{bmatrix} 0.1212 \\ 0.5369 \\ 1 \end{bmatrix}$$

$$AX_5 = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix} \begin{bmatrix} 0.1212 \\ 0.5369 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.7321 \\ 5.4378 \\ 12.027 \end{bmatrix} = 12.027 \begin{bmatrix} 0.0608 \\ 0.4521 \\ 1 \end{bmatrix}$$

$$4. \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$Ax_6 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix}$$

$$Ax_7 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 0 \\ 0.5 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0 \\ 0.2 \end{bmatrix}$$

$$Ax_8 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1.2 \\ 0.4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.6 \\ 0.2 \end{bmatrix}$$

$$Ax_9 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.6 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 0 \\ 1 \end{bmatrix} = 2.6 \begin{bmatrix} 1 \\ 0 \\ 0.3846 \end{bmatrix}$$

$$Ax_{10} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0.3846 \end{bmatrix} = \begin{bmatrix} 2 \\ -1.3846 \\ 0.7692 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.6923 \\ 0.3846 \end{bmatrix}$$

$$Ax_{11} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & -2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.6923 \\ 0.3846 \end{bmatrix} = \begin{bmatrix} 2.6923 \\ 0 \\ 1.4615 \end{bmatrix} = 2.6923 \begin{bmatrix} 1 \\ 0 \\ 0.5428 \end{bmatrix}$$